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Air Pollution, Behavioral Bias, and the Disposition Effect in China

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1 Big picture

- **Question.** Does air pollution intensify cognitive bias in financial markets, specifically the disposition effect among retail mutual fund investors?
- **Setting.** The paper uses proprietary trading records from a large Chinese mutual fund family: more than 773,198 valid accounts, seven equity funds, 247 cities, and the 2007–2015 period.
- **Core outcome.** The disposition effect is the probability of selling winners minus the probability of selling losers, aggregated mainly at the city–day level.
- **Main explanatory variable.** Air pollution is measured by the Chinese Air Quality Index (AQI). The main regression uses $\log(AQI)$.
- **Main result.** Higher AQI is associated with significantly larger disposition effects. In the baseline regressions, the coefficient on $\log(AQI)$ is positive and statistically significant across specifications.
- **Economic cost.** Investors in high-AQI/high-disposition city-days trade worse than investors in low-AQI/low-disposition city-days, with economically meaningful abnormal return differences.
- **Causal evidence.** Two identification strategies support causality: sudden pollution dissipation caused by strong wind, and the Huai River heating policy that creates a discontinuity in pollution north versus south of the river.
- **Mechanism.** The paper argues that air pollution affects mood and cognition, triggering mood regulation through realization of gains. The evidence is indirect but consistent with investors selling winners and realizing larger gains on polluted days.
- **Normative implication.** Pollution has broader economic costs than health and productivity alone. It can distort household financial decisions and redistribute wealth through trading mistakes.

2 Incremental contribution of the paper

- **Environmental economics to behavioral finance.** Prior pollution studies emphasize health, mortality, labor supply, productivity, education, and crime. This paper extends the cost of pollution into financial cognition and investor trading mistakes.
- **Micro foundation for pollution and markets.** Earlier work links weather or air quality to aggregate market returns. This paper uses investor-level mutual fund trading records to show a micro behavioral channel: polluted air increases the disposition effect.
- **Large-scale account-level evidence.** The sample covers 773,198 accounts across 247 Chinese cities, allowing city-day measurement of both investor behavior and pollution exposure.
- **Two quasi-experimental designs.** The paper does not rely only on correlations. Strong winds create short-run variation in pollution, and the Huai River heating policy creates cross-sectional RD/IV variation.
- **Mechanism inside the disposition effect.** The paper separates broad disposition effects from sign and magnitude realization preferences, connecting air pollution to mood-regulation behavior rather than merely documenting more trading.
- **Behavioral-finance extension.** The results show that seemingly nonfinancial environmental states can intensify canonical behavioral biases, reinforcing the idea that investor biases are shaped by social and physical context.

3 Data and measurement

3.1 Investor data

- The data come from a confidential Chinese mutual fund family headquartered in Shanghai.
- Each account is identified by a national identity number; investors can trade multiple equity funds through online, brokerage, or bank channels.
- For each investor, the authors observe demographics, location, account-opening information, detailed trading histories, and dividends.
- The sample contains seven equity funds and more than 773,198 valid accounts from 247 cities in mainland China.

Interpretation: The data are unusually strong for behavioral finance because they identify both the investor’s trading history and the local environment in which the investor is located.

3.2 Air pollution data

- The main air quality variable is AQI, which combines harmful pollutants such as SO_2 , NO_2 , CO , O_3 , and particulate matter.
- AQI above 100 is generally classified as unhealthy, and the paper uses daily city-level AQI data.
- The baseline specification uses $\log(AQI)$.
- Figure 2 shows that pollution was a persistent problem during the sample, especially in Beijing and other northern cities.

Interpretation: AQI is city-level and daily, which matches the aggregation of trading behavior at the city-day level.

3.3 Disposition effect measurement

For each investor–fund–day, the authors compare the current fund price to the investor’s cost basis from the full trading history. This defines whether the position is a potential winner or loser. At the city–fund–day level,

$$PSW_{jft} = \frac{\#\{\text{investors in city } j \text{ selling winner fund } f \text{ on day } t\}}{\#\{\text{investors in city } j \text{ with winner fund } f \text{ on day } t\}},$$
$$PSL_{jft} = \frac{\#\{\text{investors in city } j \text{ selling loser fund } f \text{ on day } t\}}{\#\{\text{investors in city } j \text{ with loser fund } f \text{ on day } t\}},$$

and

$$Disp_{jft} = PSW_{jft} - PSL_{jft}.$$

The authors also aggregate across funds to define $Disp_{jt}$.

Interpretation: A positive value means investors are more likely to sell winners than losers. It is a direct city-day measure of the disposition effect.

3.4 Controls and heterogeneity variables

- Regional controls include log GDP, log population, number of domestic firms, and local government income.
- Weather controls include sunshine in robustness tests.
- Investor-composition variables include older investors, female investors, migrants, education, and investment experience.
- Pollution composition is captured with indicators for whether PM2.5/PM10 is the primary pollutant.

Interpretation: These controls help separate the direct impact of air pollution from regional development, demographics, or local institutional differences.

4 Models and empirical specifications

4.1 Baseline city-day specification

The baseline panel regression can be written as

$$Disp_{jt} = \alpha + \beta \log(AQI_{jt}) + \gamma' X_{jt} + \delta_t + \theta_j + \varepsilon_{jt},$$

where δ_t are time fixed effects and θ_j are city fixed effects.

Interpretation: The coefficient β measures whether investors in the same city display a stronger disposition effect on more polluted days, relative to common time shocks and observable city controls.

4.2 Portfolio sorting performance test

Each day, cities are double-sorted into terciles by AQI and disposition effect. The authors construct buy and sell portfolios from each group and compute next-day trading performance.

$$Return_{g,t+1}^{trade} = Return_{g,t+1}^{buy} - Return_{g,t+1}^{sell}.$$

Interpretation: If the AQI-related disposition effect is a mistake, high-AQI/high-disposition trades should underperform low-AQI/low-disposition trades.

4.3 Difference-in-differences from large AQI drops and strong wind

For abrupt pollution dissipation, the DID specification is

$$Disp_{jt} = \rho_0 + \rho_1 Treated_{jt} + \rho_2 (Treated_{jt} \times After_{jt}) + \rho_3 After_{jt} + \rho_4' X_{jt} + \delta_t + \theta_j + \varepsilon_{jt}.$$

The key coefficient is ρ_2 .

Interpretation: When strong winds remove pollution from treated cities, the disposition effect should decline in treated cities relative to control cities. A negative ρ_2 supports a short-run causal effect of pollution.

4.4 Huai River regression discontinuity and IV

The first-stage RD relation is

$$AQI_{jt} = \beta_0 + \beta_1 D(North_j) + f(R_j) + \beta'_2 X_{jt} + \delta_t + \varepsilon_{jt},$$

and the reduced-form behavioral equation is

$$Disp_{jt} = \theta_0 + \theta_1 D(North_j) + f(R_j) + \theta'_2 X_{jt} + \delta_t + \varepsilon_{jt}.$$

The IV second stage uses fitted $\widehat{AQI}_{jt}$:

$$Disp_{jt} = \gamma_0 + \gamma_1 \widehat{AQI}_{jt} + f(R_j) + \gamma'_2 X_{jt} + \delta_t + \varepsilon_{jt}.$$

Interpretation: North of the Huai River, government-provided coal heating raises pollution in winter. If the discontinuity changes trading bias only when pollution is generated by heating, omitted city differences become less plausible.

4.5 Account-level robustness and mechanism tests

At the account-year level, the paper estimates annual disposition effects as a function of annual average AQI. It also uses Cox proportional hazard models:

$$h_i(t) = \gamma(t) \exp\{\beta_1 Gain_{i,t-1} + \beta_2 Gain_{i,t-1} \log(AQI_t) + \beta_3 \log(AQI_t)\}.$$

Mechanism tests interact pollution with past momentum and return magnitudes.

Interpretation: Account-level tests verify that the city-level findings are not an aggregation artifact. Mechanism tests ask whether polluted-day selling resembles mood regulation through realization of gains.

5 Identification and empirical design

5.1 Baseline concern

Air pollution is not randomly assigned. Polluted cities may differ in income, industrial structure, trading culture, migration, or investor sophistication. A simple OLS relation between AQI and disposition effects could therefore be confounded.

5.2 Why strong wind helps

- Strong wind can abruptly dissipate air pollution.
- Wind events are plausibly unrelated to mutual fund fundamentals.
- The design compares treated and control cities before and after the wind-induced pollution drop.
- Placebo tests using strong wind without pollution help rule out wind itself as the driver.

Interpretation: This design is closest to a short-run natural experiment: pollution changes while local investors, assets, and city characteristics remain fixed.

5.3 Why the Huai River helps

- The Huai River/Qinling Mountains line separates cities eligible for coal-based winter heating from cities south of the line.
- The policy increases pollution north of the river for reasons unrelated to local financial trading.
- The RD compares nearby cities just north and south of the river.
- Seasonality tests show effects in heating seasons but not nonheating seasons.

Interpretation: The heating-season pattern is especially important because permanent north-south city differences would operate throughout the year, not only during winter heating.

5.4 Remaining limitations

- Mood regulation is inferred indirectly; the paper does not observe investors' real-time moods or cognition.
- AQI is city-level, so individual indoor pollution exposure is measured with error.
- The data come from one large fund family, which raises external-validity questions.
- The setting is China, where capital-gains tax treatment and mutual fund investor behavior differ from the United States.

6 Tables and figures

6.1 Figure 1: Locations of cities and the Huai River in China + Figure 2: AQI in recent years in China

Read:

- Figure 1 plots the geographic distribution of sample cities and the Huai River/Qinling line used for the RD design.
- The wide geographic coverage is essential because the authors need variation in both pollution and investor behavior across cities.
- Figure 2 shows average AQI and confidence intervals for all sample cities and for Beijing from 2007 to 2015.
- Beijing begins with unhealthy air quality, improves around the Olympic cycle, and then worsens again, with high upper-tail pollution.

Economic message: These figures establish the institutional setting: pollution is spatially and temporally meaningful, and the Huai River line is a plausible policy discontinuity.



Fig. 1. Locations of cities and the Huai River in China. The figure plots the geographic location of the cities covered in our sample in China. Each city is represented by one dot on the map. The line in the middle of the map is the Huai River augmented by the Qinling Mountains, which geographically divide China into its southern and northern parts.

(a) Figure 1: Locations of cities and Huai River

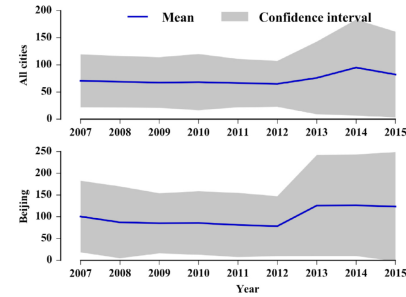


Fig. 2. AQI in recent years in China. The figure plots the mean and 90% confidence interval of the AQI for all cities in our sample (top) and those for Beijing (bottom) for the period from 2007 to 2015.

(b) Figure 2: AQI in recent years in China

6.2 Table 1: Summary statistics + Table 2: Baseline AQI-disposition relation

Read:

- Table 1 shows 144,820 city-day disposition observations, a mean disposition effect of 0.198%, PSW of 0.382%, PSL of 0.184%, and mean AQI of 80.265.
- The correlation matrix shows AQI is positively correlated with the disposition effect, but the correlation is modest, motivating panel regressions.
- Table 2 Panel A shows that high-AQI/high-disposition city-days are much more common than chance and that these trades underperform low-AQI/low-disposition trades.
- Table 2 Panel B reports positive and significant $\log(AQI)$ coefficients around 0.037–0.044 across baseline specifications.

Economic message: The baseline result is not only statistical. High-pollution/high-bias portfolios suffer negative trading performance, so the AQI-related disposition effect behaves like a costly trading mistake.

Table 1

Summary statistics.

This table presents summary statistics of the data from 2007 to 2015 used in this paper. Panel A reports numbers of observations, means, and standard deviations, along with the 5%, 25%, 50%, 75%, and 95% quantile values of the main variables, including measures of the city-level disposition effect in A1, the air quality index (AQI) in A2, and time-varying regional control variables in A3. Panel B presents the Spearman rank correlation coefficients of the variables. Coefficients that are significant at the 5% level are highlighted in bold.

Panel A: Summary statistics of main variables								
	N	Mean	Std dev	5%	25%	Median	75%	95%
A1: City-level disposition effect (city-day observations)								
Disposition effect.%	144,820	0.198	1.535	-0.662	0.000	0.000	0.000	1.867
PSW.%	144,820	0.382	1.376	0.000	0.000	0.000	0.125	2.083
PSL.%	144,820	0.184	0.857	0.000	0.000	0.000	0.011	0.952
A2: City-level air quality index								
AQI	144,239	80.265	44.250	34	54	70	94	159
A3: Time-varying local control variables								
Log_GDP	1540	15.890	1.168	14.244	15.057	15.742	16.624	18.019
Log_pop	1532	4.873	0.839	3.649	4.320	4.805	5.387	6.333
Log_num_domestic_firm	1532	5.733	1.325	3.691	4.852	5.684	6.475	7.965
Log_gov_income	1538	13.382	1.355	11.220	12.530	13.310	14.208	15.696
Panel B: The correlation matrix								
	PSW	PSL	Disposition effect	Log_GDP	Log_pop	Log_num_domestic_firm	Log_gov_income	AQI
PSW	1							
PSL	0.1153	1						
Disposition effect	0.8323	-0.4548	1					
Log_GDP	0.0063	-0.0082	0.012	1				
Log_pop	-0.0042	-0.0133	0.0036	0.8473	1			
Log_num_domestic_firm	0.0016	-0.0083	0.006	0.8268	0.7788	1		
Log_gov_income	0.0133	-0.0038	0.0141	0.902	0.77	0.7756	1	
AQI	0.0037	-0.0063	0.0068	0.0063	0.0256	0.0051	0.0171	1

(a) Table 1: Summary statistics

Table 2

The impact of air quality on trading bias: baseline analysis.

This table presents the baseline relationship between AQI and the disposition effect in regression and portfolio analysis. More explicitly, Panel A tabulates the results for portfolio analysis. For each day t during our sample period, we independently double-sort all cities into nine groups, according to terciles of AQI (high, mid, low) and those of the disposition effect (high, mid, low) and then assess the trading performance of investors in these sorted groups. Panel A1 tabulates the average value of AQI and the disposition effect in each tercile as well as the proportion of observations that falls into each group. In Panel A2, we first aggregate all buy and sell trades by investors on day t in each of the nine groups to construct their buy and sell portfolios. We then compute their trading performance as the returns generated by the buy portfolio on day $t+1$ minus the returns of the sell portfolio on the same date. Such trading performance is further adjusted based on the CAPM model, Fama-French three-factor models, and the fund's benchmark. Panel B2 then reports trading performance for investors located in Low-Low cities (i.e., cities in the bottom tercile of AQI and the disposition effect) and those in High-High cities (i.e., cities in top tercile of AQI and the disposition effect), along with the difference between the two (denoted by High-High minus Low-Low). Panel B examines the following panel specification with city- and time-fixed effects: $\text{Trading}_{it} = \alpha_0 + \alpha_1 \times \text{AQI}_{it} + \alpha_2 \times \text{X}_{it} + \delta_i + \delta_t + \epsilon_{it}$, where AQI_{it} is the air quality index value for city i on day t , Trading_{it} denotes disposition effect, and the vector X_{it} stacks a list of region-level control variables, including the regional gross domestic product (Log_GDP), total population in the region (Log_pop), the number of domestic firms (Log_num_domfirm), and local government revenue (Log_gov_income). Model (3) further controls for sunshine conditions in each city. Model (4) excludes dates with major political events (such as large party meetings) and four tier-one cities (Beijing, Shanghai, Guangzhou, and Shenzhen). The sample period is from the year 2007 to 2015. Appendix A provides more detailed variable definitions. Robust t -statistics are reported in parentheses and are based on standard errors clustered by city and date. Superscripts of *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel A: Portfolio analysis based on double sorting (AQI and disposition effect)				
A1: Tercile values of AQI(disposition effect) in parenthesis and the fraction of observation in each double-sorted group				
	Disp_Low (-0.407%)	Disp_Mid (0.020%)	Disp_High (0.977%)	
AQI_Low (49.439)	22.56%	5.08%	5.68%	
AQI_Mid (74.573)	5.96%	20.37%	6.99%	
AQI_High (116.622)	4.81%	7.86%	20.69%	
A2: Trading performance of High-High and Low-Low AQI-associated disposition groups				
	(1) Raw return(bp)	(2) Market-adjusted return(bp)	(3) 3-factor-model-adjusted return(bp)	(4) Benchmark-adjusted return(bp)
Low-Low	0.670 (0.51)	0.901 (1.33)	1.773 (2.98)**	3.784 (2.48)*
High-High	-5.987 (-5.82)**	-0.823 (-1.70)*	0.999 (0.76)	0.026 (0.02)
High-High minus Low-Low	-6.657 (4.03)**	-1.724 (2.08)*	-1.374 (1.71)	-3.758 (2.00)*
Panel B: Disposition effect regressed on Log(AQI)				
	(1) Full sample	(2) Full sample	(3) With sunshine	(4) Excluding big events and cities
Log_AQI	0.037*** (3.91)	0.037*** (3.85)	0.044*** (3.86)	0.044*** (4.30)
Log_GDP		-0.068* (-1.73)	-0.068* (-1.73)	-0.057 (-1.46)
Log_pop		0.002 (0.99)	0.003 (1.03)	0.023 (0.65)
Log_num_domestic_firm		0.036 (1.62)	0.035 (1.60)	0.044* (1.84)
Log_gov_income		0.035* (1.66)	0.035* (1.68)	0.040* (1.86)
Sunshine		0.000 (0.00)	0.000 (0.96)	
Constant	0.151*** (3.28)	0.373 (0.65)	0.364 (0.63)	0.125 (0.21)
Fixed effects and clustering	City, day of the week, month of the year, and year fixed effects; S.E. clustered by city-day			
No. of obs	144,238	144,238	144,238	128,322
R-squared	0.02	0.02	0.02	0.02

(b) Table 2: Baseline analysis

6.3 Table 3: DID on AQI drops and strong wind

Read:

- Panel A uses large AQI drops. Treated cities experience large reductions in AQI and a significant fall in disposition effect relative to controls.
- The key Treated×After coefficient in the disposition regression is negative, about -0.234 in the main specifications.
- Panel B uses strong wind with prior pollution as the treatment. The disposition effect falls when wind dissipates pollution.
- Placebo strong-wind tests without pollution do not show the same pattern.

Economic message: Table 3 is the first causal design. It shows that removing pollution quickly reduces trading bias, making persistent city-level omitted variables unlikely to explain the baseline relation.

Table 3

DID on AQI drops.

The table presents the results of two versions of difference-in-difference tests associated with drastic AQI drops. In Panel A, we first identify the treatment group by focusing on cities that have experienced (1) air pollution at the beginning of the week (i.e. an AQI above 100 before a drastic AQI drop) and (2) the treatment event of a drastic AQI drop (i.e. larger than two standard deviations) on Wednesday or Thursday. For each city in the treatment group, we identify as control group cities those that 1) have similar degrees of pollution at the beginning of a week (i.e. an AQI difference smaller than 30) and 2) do not experience abrupt AQI changes on Wednesday/Thursday (i.e. AQI changes less than one standard deviation). Panel A1 tabulates the level of AQI for the treatment and control groups before and after the treatment effect. Panel A2 presents the results of the following multivariate specification:

$$Disp_{ijt} = \rho_0 + \rho_1 \times Treated_{ijt} + \rho_2 \times Treated_{ijt} \times After_{ijt} + \rho_3 \times After_{ijt} + \rho_4 \times X_{ijt} + \delta_t + \theta_j + \epsilon_{ijt},$$

where $Disp_{ijt}$ is the disposition effect of all investors in city j on day t . $Treated_{ijt}$ is a dummy variable that takes a value of one if city j on day t is in the treatment group, and $After_{ijt}$ is a dummy variable that takes a value of one if day t is in the posttreatment period and zero in the pretreatment period. The vector X_{ijt} contains region-level control variables. In Panel B, we identify the treatment group as cities that have experienced (1) air pollution at the beginning of a week as in Panel A and (2) the treatment event of strong wind on Wednesday or Thursday (wind speed > 5 m/s). The control group is identified similar to that above. We then apply the same multivariate specification to these two samples of city-level observations. Finally, in Panel C, we identify the treatment group as cities that have experienced 1) no air pollution at the beginning of a week (i.e. AQI < 100) and 2) the treatment event of strong wind on Wednesday or Thursday (wind speed > 5 m/s), and we apply the same multivariate specification as a placebo test in Panel B. All specifications include city and time-fixed effects, with standard errors clustered at the city level. Robust t -statistics are reported in parentheses and are based on standard errors clustered by city and date. Superscripts of *, **, and *** indicate significance levels.

Panel A: DID test using large drops in AQI as the treatment event

A1: Univariate analysis				
AQI		Before event	After event	After-before
Treated		165.92	84.09	-80.93***
Control		156.8	153.03	-3.77
Treated-Control		9.12	-68.04***	-77.16*** (-24.71)
Disposition				
Treated		0.348	0.084	-0.264**
Control		0.301	0.278	-0.023
Treated-Control		0.047	-0.194**	-0.241** (-2.49)
A2: Multivariate analysis on disposition effect and investor composition				
	(1)	(2)	(3)	(4)
	y = Disposition effect		y = Log(Trading vol)	y = Fraction_HighDisp
Treated-After	-0.234** (-2.29)	-0.234** (-2.29)	-0.051 (-0.25)	0.024 (0.53)
Treated	0.137* (1.68)	0.136 (1.65)	-0.250 (-0.86)	0.056 (1.16)
After	0.121 (1.44)	0.121 (1.44)	-0.073 (-0.41)	-0.011 (-0.30)
Log_GDP	-0.227 (-1.54)	-0.369* (-1.83)	-0.399 (-0.33)	-0.396 (-1.67)
Log_pop	0.025 (0.26)	-0.015 (-0.14)	0.276 (0.30)	0.481* (1.71)
Log_num_domestic_firm		0.071 (1.17)	-0.046 (-0.20)	-0.097* (-1.81)
Log_gov_income		0.003 (0.04)	0.448* (2.10)	0.000 (0.00)
Constant	4.186 (1.56)	5.278* (1.67)	8.821 (0.64)	5.158 (1.52)
Time and city FE	Yes	Yes	Yes	Yes
Observations	2740	2740	2740	2740
R-squared	0.15	0.16	0.58	0.27

(continued on next page)

Table 3

(continued)

Panel B: DID test using strong wind as the treatment event				
y = Disposition effect	(1)	(2)	(3)	(4)
	Strong wind with pollution		Placebo tests: strong wind without pollution	
Treated-After	-0.384*** (-2.41)	-0.391*** (-2.85)	0.011 (0.15)	0.013 (0.17)
Treated	0.245 (1.53)	0.232 (1.58)	-0.006 (-0.09)	-0.003 (-0.05)
After	0.11 (0.93)	0.113 (1.38)	-0.061 (-1.10)	-0.061 (-1.10)
Log_GDP	-2.295 (-1.29)	-2.097 (-0.92)	-0.274 (-1.03)	-0.482* (-1.92)
Log_pop	0.173 (0.18)	0.206 (0.21)	0.457** (2.28)	0.183 (0.90)
Log_num_domestic_firm		0.004 (0.02)	0.333* (1.82)	0.333* (1.82)
Log_gov_income		-0.288** (-2.40)	0.200 (1.36)	0.200 (1.36)
Constant	34.579 (1.45)	35.009 (1.09)	2.551 (0.70)	2.562 (0.79)
Time and city FE	Yes	Yes	Yes	Yes
Observations	1522	1522	13,284	13,284
R-squared	0.17	0.18	0.07	0.07

(a) Table 3 Panel A: DID using large AQI drops

(b) Table 3 Panel B: DID using strong wind

6.4 Table 4: Additional DID robustness checks

Read:

- Panel A uses strong wind as an instrument for AQI and shows positive second-stage effects of instrumented pollution on disposition.
- Panel B checks alternative thresholds and restrictions for AQI drops and wind events.
- Panel C tests reverse cases of AQI increases and weak winds, finding effects consistent with the pollution interpretation.
- Robustness remains after changing treatment thresholds, excluding event days, limiting pre-event AQI gaps, and using placebo wind conditions.

Economic message: Table 4 demonstrates that the DID result is not an artifact of a single event definition. The direction of effects reverses when pollution increases, reinforcing the interpretation.

Table 4

Additional analysis related to the DID test.

This table provides various robustness checks of tests related to drastic AQI drops. Panel A examines an alternative specification in which strong wind is used as an instrument to proxy for exogenous variations of AQI. More specifically, we estimate the following specifications: 1st stage: $AQI_{i,t} = b_1 + D(Strong Wind)_{i,t} + b_2 \times X_{i,t} + \eta_{i,t}$, and 2nd stage: $Disp_{i,t} = \alpha + \beta \times AQI_{i,t} + \gamma \times X_{i,t} + \epsilon_{i,t}$, where $D(Strong Wind)_{i,t}$ is a dummy variable that takes the value of one if a strong wind as defined in Table 1 occurs in for city i on day t , and $AQI_{i,t}$ is the projected value of $ln(AQI)$ based on the first stage regressor. Other specifications are the same as in Model (2) from Panel B in Table 2. Next, Panel B1 includes similar robustness checks that apply the following new thresholds to the DID test (i.e., as seen in Panel A of Table 3): (1) the treatment group is identified based on drastic AQI drop of more than three standard deviations (instead of two standard deviations); (2) pretreatment AQI is required to be higher than 180 (instead of 100); (3) information about the treatment event day—i.e., the date when drastic AQI drops occur—in computing the posttreatment disposition effect is excluded (instead of including the event-day disposition effect); and (4) a maximum of 50 for the AQI gap between the control group and the treatment group is allowed in the first period of the week when we identify the control group (instead of capping the difference at 30). Panel B2 includes similar robustness checks when we apply these new thresholds to the second version of the DID test (i.e., as we have seen in Panel B of Table 3), except for Model (1) in which we require strong wind to have a speed of more than 7 m/s (instead of more than 5 m/s). Panel C explores the reverse scenario of Table 3, with AQI mild at the beginning of the week and then increasing drastically in general or due to a lack of wind in particular (i.e., weak wind). To conduct a test of such treatment events, we first identify the treatment group by focusing on cities that have experienced (1) little or mild air pollution at the beginning of a week (i.e., AQI < 150 on Monday and Tuesday) and (2) a treatment event of drastic AQI increases (i.e., larger than two standard deviations) or weak wind (wind speed < 2 m/s among the AQI increases sample) on Wednesday or Thursday. For each city in the treatment group, we identify as a control group cities that (1) have a similar degree of pollution at the beginning of a week (i.e., an AQI difference smaller than 30) and (2) do not experience abrupt AQI changes on Wednesday/Thursday (i.e., AQI changes less than one standard deviation). Robust t-statistics are reported in parentheses and are based on standard errors clustered by city and date. Superscripts of *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel A: Alternative specifications (strong wind as an instrument)				
	(1) 1st-stage Log(AQI)	(2) 2nd-stage Disposition	(3) 1st-stage Log(AQI)	(4) 2nd-stage Disposition
D(Strong wind)	-0.128*** (-56.93)		-0.128*** (-56.88)	
Log_AQI_Hat		0.094** (2.08)		0.097** (2.13)
Log_GDP			-0.051*** (-4.79)	0.070*** (2.81)
Log_pop			0.055*** (6.03)	-0.039* (-1.68)
Log_num_domestic_firm			0.042*** (4.40)	-0.027* (-1.82)
Log_gov_income			-0.044*** (-7.07)	-0.018 (-1.23)
Sunshine			0.000 (0.49)	-0.000 (-0.89)
Constant	4.486*** (341.54)	-0.604*** (-2.93)	5.361*** (36.12)	-1.124*** (-2.60)
Fixed effects	City, day of the week, month of the year, and year fixed effects; S.E. clustered by city-day			
Observations	144,238	144,238	144,238	144,238
R-squared	0.27	0.02	0.27	0.02

Panel B1: Robustness checks of the DID test (abrupt drops in AQI)				
	(1) Change in AQI3sd	(2) AQI(-1) >= 180	(3) Excluding event day	(4) Pre-event AQI Diff < 50
Treated*After	-0.582*** (-2.89)	-0.481*** (-2.74)	-0.248*** (-2.57)	-0.256** (-2.58)
Treated	0.185 (0.95)	0.187 (1.60)	0.239*** (2.82)	0.140* (1.68)
After	0.248* (1.74)	0.314** (2.05)	0.071 (1.01)	0.140* (1.70)
Log_GDP	1.154 (1.10)	0.197 (0.39)	-0.775*** (-2.84)	-0.445** (-2.11)
Log_pop	-0.113 (-0.10)	0.095 (0.13)	0.036 (0.24)	0.066 (0.84)
Log_num_domestic_firm	0.251 (0.30)	-0.592 (-1.20)	0.126 (1.60)	0.071 (1.34)
Log_gov_income	-3.130*** (-2.01)	-0.875*** (-2.65)	-0.173 (-1.14)	0.001 (0.01)
Constant	21.243*** (2.42)	10.925 (1.46)	13.247*** (3.18)	6.112** (1.98)
Time and city FE	Yes	Yes	Yes	Yes
Observations	879	1233	2221	3160
R-squared	0.20	0.17	0.17	0.14

(continued on next page)

(a) Table 4 Part 1: Additional DID analysis

Table 4
(continued)

Panel B2: Robustness checks of the DID test (strong wind)				
	Wind >= 7 m/s	AQI(-1) >= 180	Excluding event day	Pre-event AQI Diff < 50
Treated*After	-0.505*** (-3.49)	-0.369*** (-2.72)	-0.357*** (-2.31)	-0.385*** (-3.88)
Treated	0.253 (1.11)	0.183 (0.76)	0.285** (2.07)	0.261*** (2.86)
After	0.272** (2.56)	0.136* (1.82)	0.067 (0.81)	0.193*** (2.89)
Log_GDP	-0.534 (-0.65)	-0.318 (-0.15)	-1.473 (-0.52)	-1.422 (-1.20)
Log_pop	0.717 (1.54)	1.232 (0.51)	-0.149 (-0.13)	0.570 (1.03)
Log_num_domestic_firm	0.173 (0.60)	-2.445 (-1.45)	0.054 (0.23)	-0.368 (-1.28)
Log_gov_income	-0.172* (-1.93)	-0.119 (-0.73)	-0.318** (-2.28)	-0.192** (-2.06)
Constant	5.497 (0.46)	14.555 (0.79)	27.125 (0.68)	23.857 (1.32)
Time and city FE	Yes	Yes	Yes	Yes
Observations	1018	762	1241	2492
R-squared	0.20	0.41	0.21	0.16

Panel C: DID test for the reverse case of AQI increases				
	Abrupt increases in AQI as the treatment event	Weak wind as the treatment event		
Treated*After	0.231*** (2.83)	0.227*** (2.81)	0.245** (2.54)	0.243** (2.53)
Treated	-0.082 (-1.12)	-0.095 (-1.29)	-0.011 (-0.09)	-0.019 (-0.16)
After	-0.057 (-0.83)	-0.055 (-0.83)	-0.040 (-0.71)	-0.047 (-0.69)
Log_GDP	-0.048 (-0.26)	0.119 (0.67)	-0.119 (-0.57)	-0.074 (-0.38)
Log_pop	-0.045 (-0.28)	0.021 (0.14)	0.085 (0.46)	0.155 (0.88)
Log_num_domestic_firm	-0.237** (-3.41)	-0.237** (-3.41)	-0.077* (-2.00)	-0.077* (-2.00)
Log_gov_income	0.125** (2.18)	0.125** (2.18)	-0.066 (-0.55)	-0.066 (-0.55)
Constant	0.450 (0.18)	-2.702 (-1.06)	0.807 (0.36)	1.046 (0.36)
Time and city FE	Yes	Yes	Yes	Yes
Observations	3683	3683	1858	1858
R-squared	0.13	0.13	0.21	0.21

(b) Table 4 Part 2: Additional DID analysis

6.5 Table 5: RD test based on Huai River policy + Figure 3: Disposition effect across Huai River

Read:

- Table 5 shows that cities north of the Huai River have higher AQI and higher disposition effects around the discontinuity.
- The discontinuity in disposition effect is about 0.19–0.21 percentage points, statistically significant under linear and quadratic specifications.
- Figure 3 visualizes the discontinuity: the fitted line jumps upward north of the Huai River under both first- and second-order polynomial fits.
- The jump is consistent with heating-induced pollution changing investor behavior rather than only general north-south differences.

Economic message: The RD design provides cross-sectional quasi-experimental evidence. It supports the claim that pollution, not just local economic development, increases behavioral bias.

Table 5

RD test based on the Huai River policy.

This table presents the results of the following test to examine the influence of the Huai River policy:

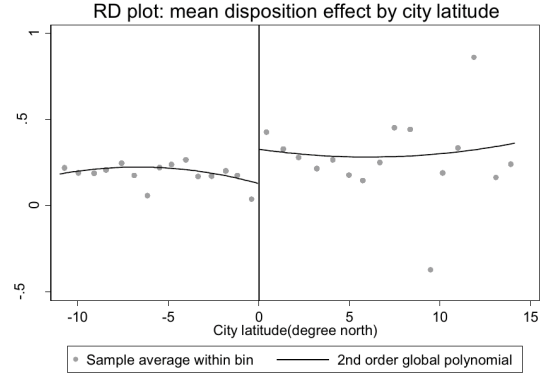
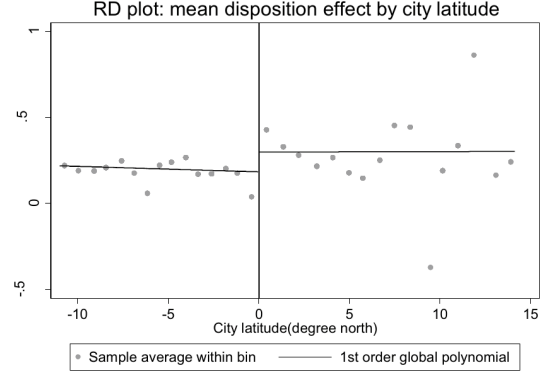
$$AQI_{ijt} = \beta_0 + \beta_1 \times D(\text{North})_{ijt} + f(R_{ijt}) + \beta_2 \times X_{ijt} + \delta_t + \epsilon_{ijt}$$

$$\text{Trading bias}_{ijt} = \theta_0 + \theta_1 \times D(\text{North})_{ijt} + f(R_{ijt}) + \theta_2 \times X_{ijt} + \delta_t + \epsilon_{ijt}$$

where AQI_{ijt} and $\text{Trading bias}_{ijt}$ measure the degree of air pollution and the disposition effect for city j in year t . $D(\text{North})_{ijt}$ is an indicator variable equal to one if city j is located north of the Huai River (augmented by the Qinling Mountains). δ_t represents the degree of northern latitude of city j relative to that of the Huai River. $f(R_{ijt})$ is parameterized as a k -order polynomial function of R_{ijt} on either side of the Huai River, and vector X_{ijt} contains a set of time-varying region-level control variables. Appendix A provides detailed definitions of all variables. Panels A and B present the results when $f(R_{ijt})$ is estimated as a linear ($k=1$) and quadratic function ($k=2$), respectively. We further require $|R_{ijt}| < 10^\circ$ in our main test. All specifications include year-fixed effect with standard errors clustered at the city level. The testing period is from 2007 to 2015. Robust t -statistics are reported in parentheses and are based on standard errors clustered by city and year. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel A: Linear specification				
	(1)	(2)	(3)	(4)
	AQI		Disposition effect	
D(North)	8.845** (2.58)	8.628*** (3.08)	0.205*** (2.86)	0.189*** (3.79)
Degree north	0.349 (0.89)	0.214 (1.12)	-0.004 (-1.15)	-0.002 (-0.40)
Log_GDP		-3.584* (-1.69)		-0.046 (-0.95)
Log_pop		14.602*** (3.13)		0.037* (1.74)
Log_num_domestic_firm		-3.267 (-1.56)		0.030 (0.72)
Log_gov_income		-0.093 (-0.06)		-0.030 (-0.46)
Constant	69.784*** (44.05)	74.399*** (3.07)	0.157*** (5.06)	0.909 (1.46)
Year fixed effect	Yes	Yes	Yes	Yes
No. of obs	709	709	709	709
R-squared	0.26	0.30	0.04	0.05
Panel B: Quadratic specification				
	(1)	(2)	(3)	(4)
	AQI		Disposition effect	
D(North)	10.741*** (2.83)	9.814*** (3.40)	0.207*** (3.01)	0.188*** (4.07)
Degree north	0.010 (0.02)	-0.049 (-0.22)	-0.005 (-1.26)	-0.002 (-0.34)
Degree north squared	-0.187*** (-4.56)	-0.163*** (-5.75)	0.000 (-0.38)	0.000 (0.20)
Log_GDP		-1.498 (-0.56)		-0.048 (-0.97)
Log_pop		12.148*** (2.79)		0.040 (1.30)
Log_num_domestic_firm		-4.184* (-1.87)		0.031 (0.68)
Log_gov_income		0.057 (0.04)		-0.030 (-0.46)
Constant	73.714*** (91.10)	59.880* (1.95)	0.160*** (4.32)	0.921 (1.38)
Year fixed effect	Yes	Yes	Yes	Yes
No. of obs	709	709	709	709
R-squared	0.29	0.32	0.04	0.05

(a) Table 5: RD test based on Huai River policy



Disposition effect across Huai River. The figure plots trading bias aggregated at the city level against degrees north of the Huai River at 33.6° (the middle of the latitude range covered by the Huai River). Each dot represents the average over each bin, and the

(b) Figure 3: Disposition effect across Huai River

6.6 Table 6: Two-stage least squares RD estimates + Table 7: Haze and investor characteristics

Read:

- Table 6 instruments AQI using the Huai River policy and finds positive significant effects of instrumented AQI on disposition effect.
- The effects are concentrated in heating seasons and are insignificant in nonheating seasons.
- Table 7 shows heterogeneous sensitivity: the pollution effect is weaker in cities with more older, better educated, or more experienced investors.
- The effect is stronger when PM2.5/PM10 is the primary pollutant.

Economic message: Table 6 strengthens causal interpretation; Table 7 helps identify who is most vulnerable and which pollutants are most relevant.

Table 6

The impact of AQI on two-stage least square RD estimations.

This table provides results of a two-stage least-square specification used to estimate the effect of AQI on investors' trading bias in the period from 2007 to 2015. The first stage is reported in Model (2) of Table 3. In the second stage, we estimate the following specification: $\text{Bias}_{it} = \beta_0 + \beta_1 \times \text{AQI}_{it} + f(R_i) + \gamma_2 \times X_{it} + \delta_i + \varepsilon_{it}$, where Bias_{it} refers to the disposition effect of all investors in city i in year t , AQI_{it} is the fitted value of AQI from the first-stage estimation, $f(R_i)$ is an indicator variable that takes a value of one if city i is located north of the Hwai River line and zero otherwise, R_i represents the degree of northern latitude of city i relative to that of the Hwai River, $f(R_i)$ is parameterized as a k -order polynomial function of R_i on either side of the Hwai River (linear in Models 1 and 2 and quadratic in Models 3 and 4), and vector X_{it} contains a set of time-varying region-level control variables. Appendix A provides detailed definitions of all variables. All specifications include year-fixed effects, with standard errors clustered at the city level. Panel A presents the results of the second-stage estimations. Panel B further splits each year into heating and nonheating seasons and reports the results of the above estimation in these two subperiods. All Craig-Donald Wald F -statistics exceed the Stock-Yogo weak instrument thresholds. Robust t -statistics are reported in parentheses and are based on standard errors clustered by city and year. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel A: Disposition effect regressed on instrumented AQI (full sample analysis)				
	(1)	(2)	(3)	(4)
	Linear specification		Quadratic specification	
AQI_hat	0.024** (2.54)	0.022** (2.08)	0.020** (2.70)	0.019** (2.17)
Degree north	-0.013 (-0.72)	-0.007 (-0.94)	-0.005 (-0.52)	-0.001 (-0.18)
Degree north squared			0.003** (2.04)	0.003** (2.22)
Log_GDP		0.036 (0.46)	-0.017 (-0.22)	0.111** (3.29)
Log_pop		-0.280*** (-3.17)	-0.191*** (-3.29)	
Log_num_domestic_firm		0.101* (1.80)	0.111* (1.70)	
Log_gov_income		-0.032 (-0.43)	-0.035 (-0.50)	
Constant	-1.499 (-1.31)	-0.707 (-0.64)	-1.283 (-1.44)	-0.217 (-0.21)
Year fixed effect	Yes	Yes	Yes	Yes
No. of obs	709	709	709	709
Panel B: Disposition effect regressed on instrumented AQI in heating vs nonheating seasons				
	Heating season		Nonheating season	
	Linear specification		Quadratic specification	
AQI_hat	0.065** (2.32)	0.015 (0.75)	0.053*** (2.80)	0.010 (0.83)
Degree north	-0.031** (-2.32)	-0.011 (-0.81)	-0.009 (-0.95)	-0.004 (-0.95)
Degree north squared			0.009** (2.22)	0.001 (0.73)
Log_GDP	0.243 (0.88)	-0.021 (-0.46)	0.049 (0.22)	-0.055** (-4.05)
Log_pop	-1.097* (-2.36)	-0.129 (-0.69)	-0.749*** (-2.60)	-0.053 (-0.62)
Log_num_domestic_firm	0.223* (1.67)	0.092 (0.91)	0.234** (2.19)	0.085 (0.90)
Log_gov_income	0.071 (0.60)	-0.044 (-0.55)	0.060 (0.58)	-0.045 (-0.58)
Constant	-4.918 (-1.31)	0.186 (0.19)	-2.947 (-1.05)	0.608* (1.88)
Year fixed effect	Yes	Yes	Yes	Yes
No. of obs	709	709	709	709
P-value of F-test		0.0274		0.0120
Heating vs. nonheating				

(a) Table 6: Two-stage least-squares RD estimates

Table 7

The impact of haze and investor characteristics: heterogeneity test.

This table explores how investors' characteristics affect the influence of air pollution on cognitive bias. In particular, we expand the baseline specification in Model 4 of Table 2 to interact AQI with a list of variables that capture the characteristics of investors in each city. *Old_High* is a dummy variable that equals one if the ratio of investors older than 40 in a city is above the median value of the ratio in the cross-section. *Female_High* is a dummy variable equal to one if the ratio of female investors in a city is higher than the median value. *Migrant_High* is a dummy variable equal to one if the ratio of migrant investors in a city is higher than the median value. *Educator_High* is a dummy variable equal to one if the ratio of more educated investors in a city (based on city census data) is higher than the median of the distribution. Following Korniotis and Kumar (2011), we classify new and experienced investors based on the number of months between the account opening date and the trading date, and we construct a dummy variable, *Experience_High*, equal to one if the ratio of experienced investors in a city is higher than the median of the distribution. *D(PM2.5/10)* is a dummy variable if the primary pollutant is PM2.5 or PM10 (more likely to penetrate into indoor environments) on day t in city i . Robust t -statistics are reported in parentheses and are based on standard errors clustered by city and date. Superscripts of *, **, and *** indicate significance levels at 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Log_AQI	0.063*** (3.77)	0.059*** (3.52)	0.024** (2.49)	0.024** (2.44)	0.038** (2.40)	0.031** (2.41)	0.080*** (4.28)	0.081*** (4.27)	0.059*** (4.24)	0.058*** (4.15)	0.029*** (2.90)	0.028*** (2.84)
Log_AQI*Old_High	-0.041*** (-2.08)	-0.035* (-1.79)										
Log_AQI*Female_High			0.067*** (2.42)	0.067** (2.41)								
Log_AQI*Migrant_High					-0.006 (-0.15)	-0.007 (-0.17)						
Log_AQI*Education_High							-0.060*** (-2.81)	-0.061*** (-2.85)				
Log_AQI*D(PM2.5/10)									-0.051*** (-2.93)	-0.050*** (-2.84)		
D(PM2.5/10)											0.010*** (3.13)	0.010*** (3.12)
Log_GDP			-0.065* (-1.68)	-0.064 (-1.84)	-0.067 (-1.20)	-0.067* (-1.72)			-0.065* (-1.66)		0.032** (1.97)	0.037 (2.02)
Log_pop			0.032 (1.00)	0.031 (0.96)	0.033 (1.19)	0.033 (1.04)			0.033 (1.03)		0.033 (1.16)	0.037 (1.16)
Log_num_domestic_firm			0.035 (1.56)	0.037* (1.67)	0.035 (0.70)	0.035 (1.57)			0.035 (1.57)		0.034 (1.55)	0.034 (1.55)
Log_gov_income			0.034 (1.59)	0.034 (1.63)	0.035 (1.09)	0.035* (1.68)			0.033 (1.57)		0.033 (1.59)	0.033 (1.59)
Sunshine			0.000 (0.96)	0.000 (0.94)	0.000 (0.58)	0.000 (0.96)			0.000 (0.96)		0.000 (0.95)	0.000 (1.02)
Constant	0.154*** (3.36)	0.367 (0.64)	0.150*** (3.25)	0.327 (0.57)	0.151 (1.63)	0.360 (0.46)	0.152*** (3.30)	0.365 (0.63)	0.154*** (3.36)	0.359 (0.62)	0.188*** (4.01)	0.453 (0.78)
Fixed effects and clustering	City, day of the week, month of the year, and year fixed effects; S.E. clustered by city-day											
No. of obs	144,238	144,238	144,238	144,238	144,238	144,238	144,238	144,238	144,238	144,238	144,238	144,238
R-squared	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02

(b) Table 7: Haze and investor characteristics

6.7 Table 8: Account-level robustness + Table 9: Selling upon momentum and counterfactual return

Read:

- Table 8 Panel A repeats the analysis at the account-year level and finds AQI positively related to individual disposition effects.
- Table 8 Panel B uses Cox proportional hazard models and shows that the gain-status selling hazard is intensified by higher AQI, especially on no-news dates.
- Table 9 shows that air pollution amplifies the tendency to sell winners after positive momentum or announcements.
- Counterfactual return tests show that winners sold on polluted days would have earned positive returns if held longer.

Economic message: These tables connect the city-level effect to individual trading and to monetary losses. Investors sell winners too early on polluted days, especially when the sale is not news-driven.

Table 8

Robustness checks conducted at the account level.

This table presents robustness checks at the account level. In Panel A, we define the annual disposition effect of an individual investor as the difference between the fraction/probability of selling winners (PSW) and that of holding onto losers (PSL) in a given year, and we link it to the annualized AQI (the average of daily values within a year) to which the investor is exposed, following the baseline regression model presented in Table 2. Robust t -statistics are reported in parentheses and are based on standard errors clustered by investor and year. In Panel B, we estimate the following Cox proportional hazards model at the account level: $\lambda(t) = \gamma(t) \times \exp(\beta_1 \times \text{Gain}_{i,t-1} + \beta_2 \times \text{Gain}_{i,t-2} + \beta_3 \times \ln(\text{AQI}_t) + \beta_4 \times \ln(\text{AQI}_{t-1}))$, where $\lambda(t)$ is the hazard function describing the selling decision of an investor since the purchase of the asset, $\gamma(t)$ is the baseline hazard, and $\text{Gain}_{i,t-1}$ is a dummy variable that takes the value of one if the underlining asset of investor i incurs capital gains on date t (and zero if it indicates capital losses). We follow the restrictions in Jovanović et al. (2005) to select a trade with which we can unambiguously match the purchase and sale dates. Specifically, we exclude sales preceded by multiple purchases and include only the first sale in our sample if a single purchase is followed by multiple sales. Models (1) and (2) are estimated based on the full sample. Models (3) and (4) are further estimated based on two subsamples according to whether the selling dates are associated with fundamental fund-level news. In particular, we classify days around fund announcements (from the announcement date to three days later) as news dates, where announcements include quarterly reports, dividends, turnovers of the management team, and changes in investment policies, etc. Other days are accordingly classified as no-news dates. In all models, we control for quarter fixed-effects. Robust t -statistics are reported in parentheses and are based on standard errors clustered by account and sale date. The sample period is from 2007 to 2015. Superscripts of *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel A: The effect of AQI on account-year disposition effect (account-level analysis for the baseline specification)				
	(1) Full Sample	(2) Full Sample	(3) With sunshine	(4) Excluding big events and cities
Log_AQI	0.066*** (2.72)	0.069*** (2.70)	0.069*** (2.73)	0.105*** (4.16)
Log_GDP		-0.224*** (-5.17)	-0.226*** (-5.22)	-0.187*** (-5.24)
Log_pop		-0.051*** (-1.52)	-0.051*** (-1.54)	-0.051*** (-1.52)
Log_num_domestic_firm		0.067*** (5.24)	0.068*** (5.29)	0.049*** (3.74)
Log_gov_income		0.016*** (1.66)	0.015*** (1.60)	0.025*** (2.86)
Sunshine			-0.002 (-1.50)	
Constant	0.449*** (4.07)	3.762*** (5.01)	3.813*** (5.08)	2.953*** (5.42)
Fixed effects and clustering		Investor, year-fixed effects (S.E. clustered by investor-year)		
No. of obs	549,211	549,211	549,211	425,403
R-squared	0.85	0.85	0.85	0.85

Panel B: The effect of AQI in Cox proportional hazards models (account-level analysis on selling decisions)				
	(1) Full Sample	(2) Subsamples of selling dates with/without news	(3) Subsamples of selling dates with/without news	(4) Subsamples of selling dates with/without news
Gain	0.298*** (54.90)	0.115** (2.34)	0.150*** (2.86)	-0.138 (-1.50)
Gain*Log(AQI)		0.043*** (3.71)	0.078*** (5.85)	0.015 (0.70)
Log(AQI)		-0.078*** (-7.46)	-0.111*** (-7.46)	-0.037*** (-1.97)
Log_GDP		-0.073*** (-11.81)	-0.074*** (-11.86)	0.003 (0.28)
Log_pop		0.027*** (5.50)	0.035*** (6.89)	-0.083*** (-8.57)
Log_num_domestic_firm		0.021*** (8.13)	0.027*** (7.10)	0.023*** (3.28)
Log_gov_income		0.035*** (8.17)	0.035*** (8.28)	0.064*** (4.56)
Fixed effects and clustering		Y	Y	Y
No. of obs	252,541	252,541	197,549	54,992

(a) Table 8: Account-level robustness

Table 9

Selling upon momentum and counterfactual return.

Panels A1 and A2 examine how air pollution affects investors' selling decision conditioning on calendar-month momentum and fund announcements. More explicitly, Models (1) and (2) present the results of the following pooled logit regressions: $D_{it} = \beta_1 \times \text{MOM}_{i,t-1} + \beta_2 \times \text{MOM}_{i,t-2} + \beta_3 \times \ln(\text{AQI}_{i,t}) + \beta_4 \times \ln(\text{AQI}_{i,t-1})$, where D_{it} denotes the dummy variable that takes the value of one if investor i sells a fund on any date t that belongs to the first ten working days (i.e., first two weeks) of a calendar month and zero otherwise (i.e., all account-fund-date observations are pooled in this regression, as long as the date of the observation belongs to the first two weeks of a calendar month); $\text{MOM}_{i,t}$ is the return of the fund in the previous month; and $\ln(\text{AQI}_{i,t})$ measures the level of air pollution faced by investor i on date t . Market return and its potential interaction with air pollution are explicitly controlled. Models (3) and (4) expand the selling decision dates to include all feasible trading dates, whereby $\text{MOM}_{i,t}$ is defined in this case as fund returns in the one-month period prior to date t . Panel A2 further applies a similar specification to the selling decision of investors after fund announcement (more specifically during the first and second ten working days of the postannouncement period) by regressing pooled selling dummy variables on the corresponding announcement-period return interacted with air pollution. Finally, for actual sales occurring during the first 10 days of a calendar month or the first 10 days of any postannouncement period, Panels B1 and B2 provide a counterfactual analysis of what investors could have earned if they had retained the asset for another 20 working days (i.e., approximately one month). More explicitly, we examine the following specification: $\text{Ret}_{i,t-1,t+20} = \beta_1 \times \text{MOM}_{i,t-1} + \beta_2 \times \text{MOM}_{i,t-2} + \beta_3 \times \ln(\text{AQI}_{i,t}) + \beta_4 \times \ln(\text{AQI}_{i,t-1})$, where $\text{Ret}_{i,t-1,t+20}$ is the counterfactual return that can be generated by the asset in a hypothetical period of 20 working days right after the actual selling date. $\text{MOM}_{i,t}$ refers to the preselling momentum in Panel B1 and announcement-period return in Panel B2, and $\ln(\text{AQI}_{i,t})$ measures air pollution of the actual selling date t . Superscripts of *, **, and *** indicate significance levels at 10%, 5%, and 1%, respectively.

A. Selling decision of investors vs. past momentum (logistic model)								
A1. Selling decision vs. past-month momentum				A2. Selling decision vs. announcement momentum				
Selling on the first 10 days of each calendar month		Selling on all dates after the announcement period		Selling on the first 10 days after the announcement period		Selling on 11–20D after the announcement period		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
Mom (last month returns)	3.949*** (140.23)	2.402*** (9.03)	3.782*** (203.42)	3.074*** (17.59)	Announcement period ret	11.691*** (36.67)	-2.088 (-1.45)	7.659*** (22.93)
Mom* AQI		0.359*** (5.80)	0.163*** (4.02)	Ann-period ret* AQI		2.759*** (8.58)		0.335 (1.01)
AQI		0.038*** (10.03)	0.076*** (18.26)	AQI		0.236*** (18.87)		0.188*** (15.15)
Lagged market ret	Yes	Yes	Yes	Yes	Ann-period market ret	Yes	Yes	Yes
Lagged market* AQI	No	Yes	No	Yes	Ann-period market* AQI	No	Yes	No
Observations	55,115,198		11,25,80,000		Observations	1,43,67,157		1,41,00,073

B. Counterfactual returns generated by winners sold by investors in polluted days (in a hypothetical 20D period right after the actual selling date)						
B1. Counterfactual 20-D return (%) vs. MOM			B2. Counterfactual 20-D return (%) vs. Announcement MOM			
(if investors do not sell in the first 10 days of a month)			(if investors do not sell in the first 10 days immediately after the announcement period)			
(1)	(2)	(3)	(4)	(5)	(6)	
Selling day AQI	1.231*** (26.79)		0.666*** (13.99)	Selling day AQI	2.782*** (26.43)	1.223*** (11.95)
Mom (prior to selling)		5.541*** (16.45)	-49.081*** (-15.86)	Ann-period ret (prior to selling)		70.322*** (20.93)
Mom* AQI			12.717*** (17.74)	Ann-period ret* AQI		65.042*** (17.88)
Constant	-6.001*** (-24.86)	-0.921*** (-4.46)	-3.865*** (-16.06)	Constant	-9.664*** (-20.92)	1.804*** (37.11)
Lagged market ret	Yes	Yes	Yes	Ann-period market ret	Yes	Yes
Lagged market* AQI	No	No	Yes	Ann-period market* AQI	No	Yes
Fund FE	Yes	Yes	Yes	Fund FE	Yes	Yes
Observations	193,989	193,989	193,989	Observations	46,241	46,241
R-squared	0.075	0.074	0.080	R-squared	0.325	0.323

(b) Table 9: Momentum and counterfactual returns

6.8 Table 10: Trading responses to past return and the influence of air pollution

Read:

- Panel A tests sign realization preference around zero returns across holding horizons.
- The baseline sign effect is strongest at short horizons but mixed at longer horizons.
- AQI interactions with the sign indicator are not consistently significant, suggesting air pollution is not mainly increasing the pure sign-realization jump.
- Panel B tests magnitude realization. AQI interacts more strongly with return magnitudes, especially for short-horizon winners.

Economic message: Table 10 is central for mechanism. The evidence is more consistent with polluted investors realizing larger gains to regulate mood than with a simple increase in any gain-realization sign effect.

7 Short summary

- The paper links environmental pollution to behavioral finance using proprietary Chinese mutual fund account data.
- It measures the disposition effect city-by-city and day-by-day as $PSW - PSL$.
- Baseline regressions show that higher AQI increases the disposition effect.
- Double sorts show that high-AQI/high-disposition trades underperform, meaning the bias has real trading costs.
- Strong-wind DID tests show that when pollution suddenly dissipates, the disposition effect falls.

Table 10

Trading responses to past return and the influence of air pollution.

This table examines realization preferences related to the disposition effect as well as how air pollution influences them. We first apply the regression discontinuity analysis of Ben-David and Hirshleifer (2012, Table 2) to the selling decision of investors for various holding horizons, when returns since purchase are in a small region around zero. Panel A focuses on the region with 0.1 standard deviations from zero with third-degree polynomials. Panel B conducts the magnitude test of Ben-David and Hirshleifer (2012; in their Table 4), in which investors' selling decisions are regressed on $Ret- = \text{Min}[0, \text{return since purchase}]$ and $Ret+ = \text{Max}[0, \text{return since purchase}]$ and a list of control variables in a probit specification. Both panels further report the influence of air pollution by interacting air pollution with the corresponding return characteristics of interest (i.e., $I[ret > 0]$ in Panel A and $Ret- / Ret+$ in Panel B). The Internet Appendix provides related summary statistics and more robustness checks for both tests. Robust t -statistics are reported in parentheses and are based on standard errors clustered by investor. Superscripts of *, **, and *** indicate significance levels at 10%, 5%, and 1%, respectively.

Panel A. Discontinuity analysis on sign realization preference (dependent variable = $I[\text{Sell}] \times 100$; range = 0.1 stdev around zero; 3rd polynomials)						
	Short-term periods (1 to 20 days) (1)	(2)	Mid-term periods (21 to 250 days) (3)	(4)	Longer periods (> 250 days) (5)	(6)
$I[ret > 0]$	0.333*** (4.21)	0.775** (2.21)	-0.017 (-0.56)	0.004 (0.05)	-0.046** (-2.02)	0.018 (0.31)
$I[ret = 0]$	-0.284*** (-5.55)	-2.526*** (-10.00)	-0.229*** (-10.90)	-0.228*** (-3.51)	-0.127*** (-7.24)	-0.022 (-0.51)
$I[ret > 0] \times \text{Logaqi}$		-0.095 (-1.19)	-0.005 (-0.24)	-0.005 (-0.24)	-0.015 (-1.20)	-0.015 (-1.20)
$I[ret = 0] \times \text{Logaqi}$		0.525*** (8.93)	-0.000 (-0.02)	-0.000 (-0.02)	-0.024*** (-2.61)	-0.024*** (-2.61)
Logaqi		-0.515*** (-8.80)	-0.004 (-0.31)	-0.004 (-0.31)	0.027*** (3.04)	0.027*** (3.04)
$\text{Sqrt}(\text{Time})$	-0.043*** (-5.47)	-0.039*** (-4.98)	-0.024*** (-19.42)	-0.024*** (-19.46)	0.002*** (4.26)	0.002*** (4.11)
Polynomials	Yes	Yes	Yes	Yes	Yes	Yes
Polynomials with $\text{sqrt}(\text{time})$	Yes	Yes	Yes	Yes	Yes	Yes
Polynomials with positive and negative indicator	Yes	Yes	Yes	Yes	Yes	Yes
Observations	963,721	963,721	1854,455	1854,455	1366,419	1366,419
R-squared	0.012	0.012	0.001	0.001	0.000	0.000

Panel B: The Ben-David and Hirshleifer (2012) magnitude test (dependent variable = $I[\text{Sell}] \times 100$)						
	Short-term periods (1 to 20 days) (1)	(2)	Mid-term periods (21 to 250 days) (3)	(4)	Longer periods (> 250 days) (5)	(6)
$Ret+$	5.108*** (58.52)	-4.237*** (-4.80)	4.749*** (86.12)	5.395*** (10.69)	2.044*** (21.97)	0.890 (1.07)
$Ret-$	1.616*** (8.96)	1.428 (0.86)	-0.376*** (-5.21)	3.269*** (4.65)	-0.596*** (-5.92)	-3.710*** (-4.13)
$I[ret > 0]$	0.090*** (7.15)	0.009 (0.14)	0.236*** (30.83)	-0.041 (-0.99)	0.050*** (3.80)	0.472*** (6.78)
$I[ret = 0]$	-0.676*** (-26.13)	-1.462*** (-5.24)	-0.812*** (-23.45)	-2.304*** (-6.58)	-0.568*** (-8.82)	-0.537 (-0.77)
$Ret+ \times \text{Logaqi}$		2.221*** (11.74)	0.605*** (5.53)	0.605*** (5.53)	0.594*** (3.24)	0.594*** (3.24)
$Ret- \times \text{Logaqi}$		-0.511 (-1.36)	-0.926*** (-6.14)	-0.926*** (-6.14)	0.199 (0.95)	0.199 (0.95)
$I[ret > 0] \times \text{Logaqi}$		-0.005 (-0.31)	0.031*** (3.50)	0.031*** (3.50)	-0.052*** (-3.35)	-0.052*** (-3.35)
$I[ret = 0] \times \text{Logaqi}$		0.130** (2.11)	0.189** (2.56)	0.189** (2.56)	0.132 (1.37)	0.132 (1.37)
Logaqi		-0.102*** (-7.56)	-0.051*** (-7.19)	-0.051*** (-7.19)	0.034*** (2.82)	0.034*** (2.82)
Control variables	Same as Table 4 in Ben-david and Hirshleifer (2012). Models (2),(4),(6) include $\text{sqrt}(\text{Time})$ and interactions.					
Observations	4357,608	4357,608	16,326,851	16,326,851	20,158,791	20,158,791
Pseudo R2	0.0321	0.0330	0.0317	0.0322	0.00948	0.0100

- Huai River RD/IV tests show higher pollution and higher disposition effects north of the river, especially during heating seasons.
- Account-level tests and Cox hazard models confirm that the effect is not solely a city aggregation artifact.
- Heterogeneity tests show that older, more educated, and more experienced investors are less sensitive to pollution.
- Mechanism tests suggest that air pollution affects magnitude realization more than sign realization.
- The paper therefore argues that air pollution-induced mood regulation may help explain the trading behavior.

8 Conclusion

8.1 Core conclusion

Li, Massa, Zhang, and Zhang show that air pollution intensifies a canonical behavioral bias: the disposition effect. Their evidence begins with strong correlations between AQI and city-level selling behavior, but it is not limited to correlations. The strong-wind DID and Huai River RD/IV designs both point to a causal effect of air pollution on investor mistakes.

8.2 Economic intuition

The intuition is that air pollution worsens mood, cognition, and mental well-being. Investors exposed to worse air quality may become more prone to behavioral shortcuts. In the setting of mutual fund trading, one

natural shortcut is to realize gains while avoiding losses. Because realized gains can provide utility bursts, polluted-day investors may sell winners too early as a form of mood regulation.

8.3 Incremental contribution restated

The paper’s incremental contribution is to show that environmental conditions can shape financial market behavior at the micro level. It moves the pollution literature beyond health and labor productivity, and it moves behavioral finance beyond stable investor traits by showing that daily environmental exposure changes a classic behavioral bias.

8.4 Why the causal designs matter

The DID design exploits abrupt pollution changes caused by strong winds, so the same cities can be compared before and after exogenous pollution dissipation. The Huai River design exploits a policy-generated discontinuity in winter heating. The consistency across these designs matters because each addresses different confounds: wind handles short-run city fixed factors, while Huai River handles cross-sectional endogeneity through policy-induced pollution.

8.5 Mechanism and limitations

The mechanism evidence is suggestive rather than definitive. The authors cannot directly observe mood, anxiety, cognition, or psychological stress at the moment of trading. Their mechanism tests infer mood regulation from return-based selling patterns, especially the tendency to sell larger gains on polluted days. This is plausible and consistent with realization-utility theory, but it is not exclusive; pollution may also affect attention, patience, self-control, or risk processing.

8.6 Takeaway

The broad takeaway is that pollution is a financial-market externality. It does not merely harm health or productivity; it can distort household financial decisions and generate trading losses. Clean-air policy may therefore improve financial welfare and reduce wealth transfers from investors most exposed to pollution to investors less exposed or less behaviorally affected.